

Devalla Pranav

devallapranav192k5@gmail.com

AI/ML intern

Role: Computer Vision

Report 1 – Enhanced 2D Virtual Apparel Fitting System

Introduction

The virtual apparel fitting system allows users to visualize garments digitally before purchase. The performance of the system depends on accurate garment placement, body understanding, and realistic rendering quality. This phase focused on improving the existing 2D virtual fitting process by refining clothing alignment, segmentation quality, pose handling, and output appearance.

Clothing Alignment Improvement

One of the important objectives was improving garment fitting accuracy on different body structures and poses. Previous outputs showed issues such as sleeve mismatch, shoulder displacement, and incorrect torso fitting. To improve this, alignment techniques and body landmark guidance methods were studied to make clothing adapt naturally to user posture. Different poses such as front standing, side orientation, and arm movement conditions were evaluated to observe fitting stability.

Segmentation and Pose Analysis

The body understanding module was enhanced to improve separation between body regions and background areas. Better region extraction helped reduce overlap between hair, hands, and garments. Pose analysis was also considered important because body posture directly influences fitting quality. Multiple key body regions were analyzed to support improved garment positioning and realistic output generation.

Visual Enhancement and Background Handling

Visual quality improvements focused on garment overlay refinement and smoother blending. Earlier outputs contained visible garment borders and unnatural transitions. Refinement methods were applied to reduce artifacts and improve realism. Background complexity was also considered because noisy

environments affect segmentation quality. Different image environments were evaluated to improve robustness.

Conclusion

The enhancement process improved garment placement quality, body understanding, visual appearance, and environmental handling. These improvements increased realism and provided better fitting performance for different user conditions.

Report 2 – Advanced 3D Virtual Fitting Research and Prototype Development

Introduction

3D virtual fitting systems provide users with an interactive experience by creating digital body representations and simulating garments on them. Compared to traditional image-based fitting systems, 3D technology focuses more on body structure, movement, and realistic cloth behavior. This research phase concentrated on studying the components required for building an advanced 3D virtual fitting environment.

Avatar Generation and Body Reconstruction

The initial stage of the study focused on creating virtual avatars that can represent user body shapes digitally. Avatar generation plays an important role because it acts as the base model for garment visualization. Different reconstruction approaches were explored to understand body structure, surface formation, and digital representation. The objective was to improve personalization and support realistic garment fitting.

Cloth Simulation and Motion Tracking

Garment simulation was another major area of research because realistic cloth movement improves user experience. The study considered cloth behavior such as folding, stretching, and movement adaptation during body motion. Motion tracking was also analyzed to understand how user movement can be transferred to virtual avatars. This helps create dynamic fitting experiences and improves interaction between users and the system.

Rendering and Personalization

Rendering quality directly affects the final visualization of virtual fitting outputs. Therefore, attention was given to improving texture appearance, surface details, and visual quality. User measurements such as body dimensions and proportions were also considered to improve personalization and fitting precision.

Conclusion

The research phase established a foundation for future implementation of a 3D virtual fitting system by studying avatar creation, cloth simulation, motion handling, and rendering methods. The work supports future development of more immersive and realistic virtual shopping experiences.

Report 3 – AI Performance Enhancement and Optimization

Introduction

Artificial Intelligence optimization plays an important role in improving the efficiency and performance of virtual fitting systems. Better optimization improves fitting accuracy, reduces execution time, and supports faster output generation. This phase focused on improving model performance by analyzing body landmark detection, segmentation quality, inference speed, and recommendation mechanisms.

Landmark Detection and Processing Improvement

Body landmark analysis is essential because garment fitting depends on accurate body understanding. The study focused on improving the detection of important body regions to support better garment placement and alignment. Different approaches were evaluated to improve pose handling and fitting precision while maintaining processing efficiency. The objective was to achieve better accuracy without increasing computational complexity.

Speed Optimization and Lightweight Processing

Another important area was improving system responsiveness. Processing speed directly affects user experience, especially in real-time virtual fitting environments. Optimization techniques were studied to reduce execution delay and improve overall efficiency. Lightweight processing methods were also

explored to support deployment on systems with limited computational resources.

Segmentation and Recommendation Enhancement

Segmentation quality influences body understanding and garment overlay performance. Therefore, improvements focused on reducing overlap issues and enhancing region extraction quality. Recommendation mechanisms were also analyzed to support personalized output generation based on user information and clothing characteristics.

Conclusion

The optimization phase improved overall system performance by focusing on body analysis, execution efficiency, segmentation quality, and recommendation processes. These improvements provide a better foundation for future development and deployment of intelligent virtual fitting systems.

Report 4 – Dataset Organization and Quality Improvement

Introduction

Datasets play an important role in developing AI-based virtual fitting systems because model performance depends on data quality and diversity. Well-organized datasets help improve learning capability, reduce errors, and increase fitting realism. This phase focused on dataset expansion, organization, and quality enhancement to support future system development.

Dataset Expansion and Diversity

The dataset preparation process concentrated on increasing clothing variations and user diversity. Different apparel categories and body structures were considered to improve representation and reduce model bias. Additional attention was given to pose diversity because virtual fitting systems must work effectively under different user positions and movements.

Dataset Organization and Annotation

A structured dataset arrangement was prepared to simplify data management and future training activities. Organizing data into different categories improves accessibility and maintenance. Annotation activities were also considered

important because properly labeled information improves model understanding and overall training quality.

Cleaning and Validation Process

Dataset cleaning focused on improving overall quality by removing unwanted samples and reducing noise. Images with low quality, duplication, or inconsistencies were reviewed to maintain reliable training data. Validation procedures were also considered to ensure better organization and support future implementation stages.

Conclusion

The dataset improvement phase enhanced data organization, quality, and diversity for virtual fitting applications. The prepared datasets provide a better foundation for future model training and support development of more accurate and realistic virtual fitting systems.

Report 5 – Virtual Fitting System Workflow Documentation

Introduction

The workflow architecture defines the sequence of operations performed within the virtual fitting system. A structured workflow improves coordination between different processing stages and supports efficient output generation. This documentation phase focused on understanding the flow of data, interaction between modules, and the complete process from input acquisition to final output generation.

Input Processing and Preparation

The workflow begins with collecting user images and garment information. Before processing, the input images undergo preparation activities such as resizing and normalization to improve consistency. Proper input preparation is important because it directly affects body analysis and garment fitting quality in later stages.

Body Analysis and Region Extraction

After preprocessing, the system analyzes body structure and posture information to understand user orientation and fitting requirements. Region extraction is then performed to separate important body areas and improve understanding of

clothing placement. This stage helps reduce overlap issues and improves fitting accuracy.

Garment Adaptation and Visualization

The garment adaptation stage aligns clothing according to user structure and body posture. After fitting operations are completed, visualization techniques are applied to improve appearance and generate realistic outputs. Refinement processes help improve output quality and provide a better user experience.

Conclusion

The workflow documentation phase defined the operational structure of the virtual fitting system and explained the interaction between major processing stages. The documented workflow provides a strong foundation for future optimization, implementation, and deployment activities.